INCH-POUND
MIL-I-24092/11
4 May 1989

MILITARY SPECIFICATION SHEET

INSULATING VARNISH, WATER BASE, BAKING, SEMI-RIGID, FOR DIP PROCESSING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24092.

REQUIREMENTS:

1. Grade - WBH: Clear, solvent containing baking varnish that cures to a semi-rigid film.

2. Class - 130 to 200. The cured varnish shall meet the requirements of the specified thermal class when determined in accordance with MIL-I-24092.

3. Composition - III. Composition III varnishes are water reducible varnishes in which the solvent consists primarily of water along with lesser amounts of organic co-solvents. These are relatively complex solvent mixtures as compared to those used in the CB and CBH grade varnishes. Composition III varnishes offer higher flash points and reduced aromatic emission.

4. The varnishes shall meet the qualification and quality conformance requirements specified in tables I and II.

AMSC N/A FSC 5970

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TABLE I. Qualification requirements, grade WBH.

Characteristics (tests)	Units	Requirements
Storage life	Months, minimum	6
Nonvolatiles	Percent	50 to 60
Flash point	°C, minimum	90
Drying time at 150°C	Minutes, maximum	100
Viscosity: 20 r/min	Centipoise	200 to 500
Build, as received	Mil, minimum	1.0
Drainage	Weight percent, maximum	25
Dielectric strength:		
96/23/50	Volts per mil, minimum	2000
96/23/96	Percent decrease from 96/23/50 value, maximum	25
24/23/water	Percent decrease from 96/23/50 value, maximum	25
Dissipation factor:		
at 23°C	Percent, maximum	0.10
at 150°C	Percent, maximum	0.50
Dielectric constant:		
at 23°C	Maximum	3.0
at 150°C	Maximum	4.0
Hardness	Shore D, minimum	85
Bond strength		
at 23°C	Pounds force, minimum	35
at 150°C	Pounds force, minimum	10

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TABLE I. Qualification requirements, grade WBH - Continued.

Characteristics (tests)	Units	Requirements
Thermal endurance:		
Class 130	Temperature index, °C, minimum	130
Class 155	Temperature index, °C, minimum	155
Class 180	Temperature index, °C, minimum	180
Class 200	Temperature index, °C, minimum	200
Salt water proofness	Hours, minimum	100
Hydrolytic stability	Percent decrease in bond strength, maximum	50
Chemical resistance:		
Hydraulic fluid	Percent decrease in bond strength, maximum	25
Lubrication oil		25
Cleaning fluid		25
Distilled water		25
Detergent solution		25

TABLE II. Quality conformance requirements, grade WBH.

Characteristics (tests)	Units	Requirements
Nonvolatiles	Percent of qualification value	± 5
Drying time at 150°C	Minutes, maximum	120
Viscosity at 20 r/min	Centipoise, percent of qualification value	± 15
Build, as received	Mil, minimum	0.9
Specific gravity	Percent of as-received value	± 2.0
Hardness	Shore D, minimum	82
Bond strength at 23°C	Pounds force, minimum	30

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NOTE:

Part or Identifying Number (PIN). Part number shall be M24092/11-(dash number indicating thermal class). For example, M24092/11-155 describes a material meeting the requirements of this specification and having class 155 thermal endurance.

Custodians:
Army - ER
Navy - SH
Air Force - 20

Review activities:
Army - MI, AR
Navy - EC, YD, OS
Air Force - 99
DLA - GS, DS

User activities:
Army - ME
Navy - MC, AS

Preparing activity:
Navy - SH
(Project 5970-0661-11)

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